

Lab 08

Programming Basys-3 Board

Name: Khuzaima Ali Khan	ID: RA	Section: T1
-------------------------	--------	-------------

8.1 Objectives

After performing this lab, students should be able to:

- To synthesize bitstream for Basys-3 FPGA board
- Program Basys-3 FPGA board for adder-subtractor design
- Get exposure to design constraint files.

8.2 Introduction

The Basys 3 board in Figure 8.1 is a complete, ready-to-use digital circuit development platform based on the latest Artix®-7 Field Programmable Gate Array (FPGA) from Xilinx®. It includes a large collection of on-board I/O devices such as, switches, LEDs, push buttons, seven segment display, and all required FPGA support components. This allows a large number of designs to be completed without the need for any additional hardware. The board is further explained in Tutorial [Video 3: Download Design to FPGA Board](#) uploaded on Canvas. It is exclusively designed for the Vivado® Design Suite, a software suite produced by Xilinx for synthesis and analysis of HDL designs, with additional features for system on a chip development and high-level synthesis.

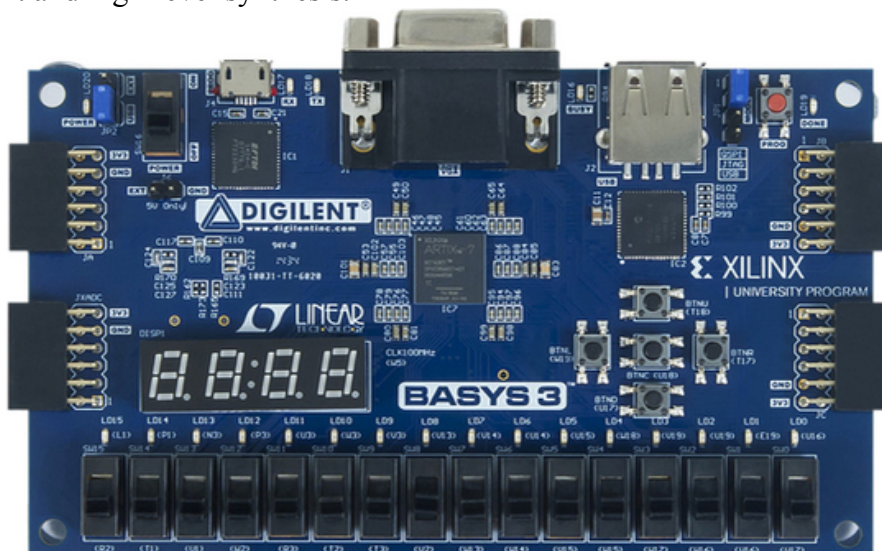


Figure 8.1: Basys-3 FPGA Board



8.3 Creating Design Modules and Testbench on Vivado Design Suite

Xilinx Vivado helps to test Verilog designs before uploading it on FPGA.

Task a)

Use *myAND* module already created in Lab 05 and test your simulation. Attach screenshot of timing diagram of AND gate in worksheet.

Reference: Follow tutorial [Video 1: Creating Project for Basys-3 board in Xilinx Vivado](#) & [Video 2: Creating Testbench and Simulate Design in Xilinx Vivado](#) in Lab 05 and simulate *and_gate* logic design on Vivado Design Suite.

8.4 Programming Basys-3 FPGA Board

Task b)

Follow video tutorial 3 given on previous page to program Basys-3 board for *and_gate* design. Attach snapshot of simulation output of AND gate in worksheet (Picture of your FPGA Board).

Demonstrate the programmed FPGA to your respective RA.

8.5 Testing Adder/Subtractor Design on Basys-3 FPGA Board

The board has 4-digit seven segment display each activates with an active low signal. As per our design requirement, we need one seven segment display. Thus, one of the displays should be activated and others should remain inactive. In your add/subtract design module, add one more 4-bit output for seven segment display activation. Send a low signal on one of them and keep rest as high.

Task c)

Follow the steps listed below or the video tutorials to program the Basys-3 board for your proposed adder/subtractor design. Consider inputs are applied using slide switches and output is displayed on a seven-segment display.

1. In Vivado 2020.1 IDE, Create New Project (You can use your project created in previous Lab).
2. Create/add design modules as source files.
3. Create/add test bench as simulation file.
4. Run the simulation to verify your design.
5. Next, run synthesis.
6. Update .xdc file according to your design.
7. Run implementation.
8. Generate bitstream and program the device.



Add your design files as source files to Vivado project. Make sure the file name is same as module name. A single file can have only one design module.



For each user-defined module instantiation in main design, double check the instanced module file is in the same directory as the main design is.

Use Basys-3 user manual (uploaded on Canvas) for pins configuration of seven segment and slide switches.

Submit the following in the worksheet for Task c:

- I. *Appropriately commented code of all design modules (Testbenches are not required).*
- II. *Constraint Files.*
- III. *Picture of your programmed FPGA Board with output displayed on 7-segment display and write the details of output cases displayed.
Add at least 3 cases – addition of inputs, subtraction of inputs with a positive result, and subtraction of inputs with a negative result.*

Demonstrate the programmed FPGA to your respective RA.



Assessment Rubric
Lab 08
Programming Basys-3 Board

Name: Khuzaima Ali Khan	ID: RA	Section: T1
--------------------------------	---------------	--------------------

Marks Distribution:

		LR2 Code	LR5 Results	LR9 Report
In - Lab	Task a	-	10	10
	Task b	-	10	
	Task c	20	10	30
Total Marks: 100		/20	/30	/40
CLO Mapping		CLO3	CLO3	CLO4

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission and Viva	/10	CLO 4

CLO	Total Points	Points Obtained
3	50	
4	50	
Total	100	

For description of different levels of the mapped rubrics, please refer to the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.



Lab Evaluation Assessment Rubrics

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/ simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/ simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/ simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/ simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR9	Report	All the in-lab tasks are not included in report.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included.	Good Summary of most of the in-lab tasks is included in report. The work is supported by figures and plots with explanation.	Detailed Summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots, and tables.
AR7	Report Submission & Viva	Most of the questions are not answered/ figures are not labelled/titles are not mentioned/ units are not mentioned. No comments are present in the code.	Some of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Few comments are stated in the code.	Majority of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Comments are stated in the code.	All the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Proper comments are stated in the code.

